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Mechanical Properties of Several Fe-Ni Meteorites

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The strength and elastic constants of meteorites are of increasing interest as predictions of meteorite impacts on earth come within the realm of possibility.[1] In addition, meteorite impacts on extraterrestrial bodies provide an excellent sampling tool for evaluation of planetary compositions and properties.[2] Fe-Ni meteorites provide a well-defined group of materials of fairly uniform composition. Iron-nickel meteorites exhibit a unique lamellar microstructure, a Widmanstätten structure, consisting of small regions with steep-iron-nickel composition gradients.[3,4] This microstructure is found in the Fe-Ni system only in meteorites, and is believed to arises as a result of slow cooling in a planetary core or other large mass.[5] Meteorites with compositions consisting of between 5 and 17% nickel in iron are termed "octahedrite," and further characterized according to the width of the Ni-poor kamacite bands; "fine," (0.2-0.5 mm) "medium," (0.5-1.3 mm) and "coarse," (1.5-3.3 mm) Many meteorites have inclusions and structures indicating that the material has been shocked at some point early in its evolution.[6]

Several Iron-nickel meteorites have been examined using Vickers and spherical indentation, x-ray fluorescence, and EBSD. Direct observation of mechanical properties in these highly structured materials provides a valuable supplement to bulk measurements, which frequently exhibit large variation in dynamic properties, even within a single sample.[4] Previous studies of the mechanical properties of a typical iron-nickel meteorite, a Diablo Canyon specimen, indicated that the strength of the composite was higher by almost an order of magnitude than values obtained from laboratory-prepared specimens.[5] Additional meteorite specimens have been examined to establish a range of error on the previously measured yield, to determine the extent to which deformation upon re-entry contributes to yield, and to establish the degree to which the strength varies as a function of microstructure.

We selected meteorites exhibiting Widmanstätten structures on several scales and also several samples without ordered structures, to examine variations across the low-nickel rgeions (kamacite) and the lathes of high-nickel material (taenite). Among these were samples from the Canyon Diablo fall (type IAB, coarse octahedrite) and several fine octahedrite meteorites, from the Muonionalusta meteorite (IVA) and Gibeon fall (IVA).

Samples exhibiting no ordered structure were examined, including a fragment from Wabar (IIIB) and iron from Laguna Manantiales (iron), to compare with the unstructured fragment of Canyon Diablo previously presented.[9] Both Canyon Diablo and Wabar usually exhibit medium octahedrite structure, suggesting that these fragments may have suffered thermal or mechanical damage during passage through the atmosphere.

Yield varied somewhat between samples, but did not show a clear difference between the structured and unstructured meteorites.

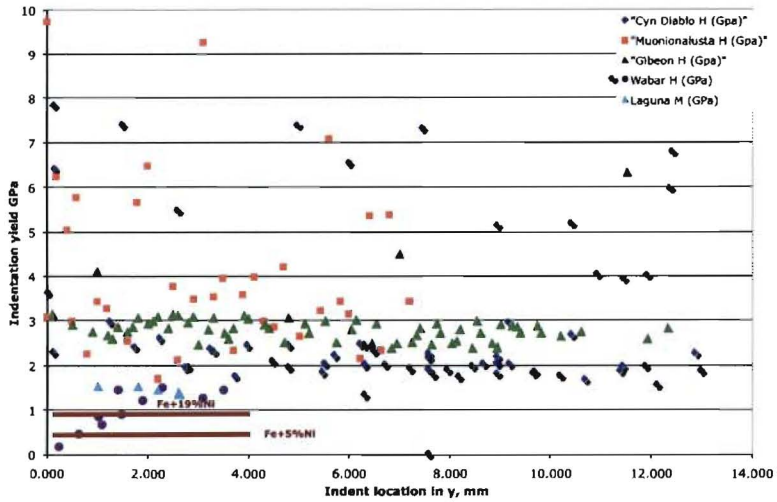


Figure 1. Yield as a function of location for several meteorites, compared to Fe-Ni alloys. Canyon Diablo, Wabar, and Laguna Manantiales are unstructured.

The Gibeon meteorite showed very little variation between properties at different locations, consistent with EDS measurements showing regions of uniform Ni concentration. Previous measurements have suggested that a strong gradient of Ni concentration exists across taenite laths.[3] The Ni was found to be uniform in a given region in our sample.

EBSD measurements of Gibeon and the unstructured Canyon Diablo both show an unusually high degree of strain within the crystallites. We attribute the high values of yield this high local strain. (Figure 3.)

The Laguna Manantiales sample appears to be large single crystal with extensive twinning. (Figure 2a) Yield of this material was nonetheless slightly higher than predicted for Fe-Ni compositions,[8] possibly as a consequence of the prevalence of twins.



Figure 2a. EBSD of Laguna Manantiales meteorite

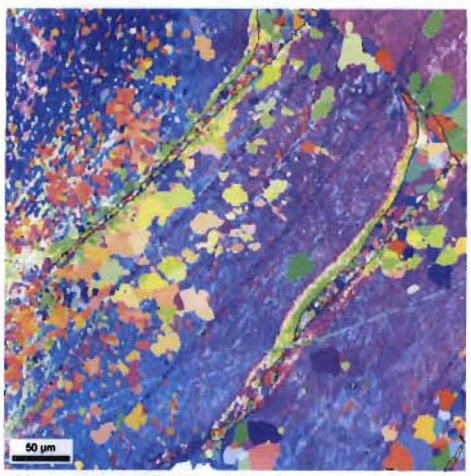


Figure 2b. EBSD of Wabar meteorite

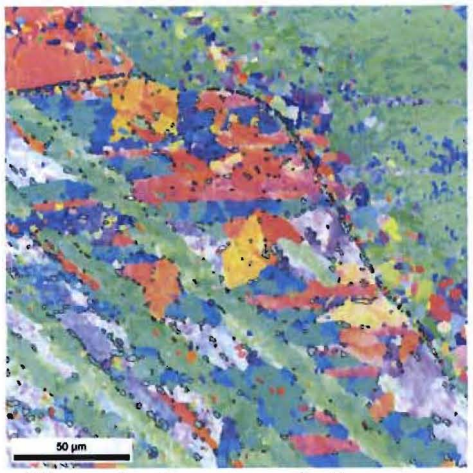
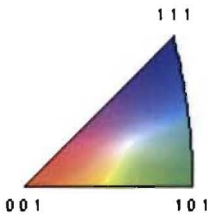


Figure 3a. Inverse pole figure, Gibeon meteorite



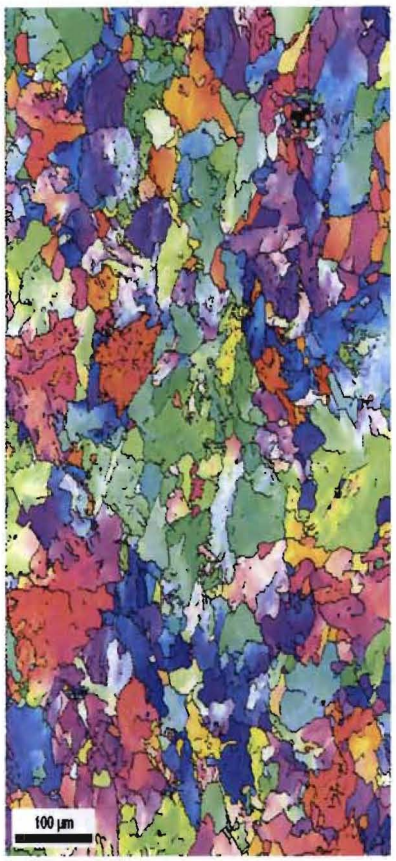


Figure 3b. Inverse pole figure, Canyon Diablo meteorite

Conclusions

Strength in all of the meteorite samples examined here was higher than predicted for iron-nickel alloys prepared in the laboratory, and higher than can be accounted for by inclusion of carbon in these alloys.[8] The unexpectedly high yield in these materials appears correlated with high localized strain in crystallites. In the case of the one non-octahedrite iron meteorite examined, the strength was less than for the octahedrites, but was high relative to iron, which may be due to extensive twinning in the meteorite sample. Inclusion of these measured values may improve atmospheric breakup and impact calculations done using iron-nickel meteorite compositions.

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